Slomins K26-SL Solar Outdoor Camera — Control4 Driver

Overview

Complete Control4 driver for Slomins K26-SL Solar-Powered Outdoor Camera with CldBus API integration, MQTT event streaming, RTSP video streaming, battery management, and wake-on-demand support.

Version: 3
Package: Slomins-outdoor-K26.c4z
Minimum Control4 OS: 3.3.2+

Purpose

Control4 driver for K26-based Tuya cameras with CldBus API integration, MQTT events, and notification handling.

Key Files

- `driver.lua` — main driver: init, auth flows, MQTT, device events, notification flow, URL generation.
- `mqtt_manager.lua` — MQTT helper for connections and message handling.
- `driver.xml` — driver manifest and properties.
- `CldBusApi/*` — helper libs: `dkjson.lua`, `http.lua`, `transport_c4.lua`, `util.lua`, `sha256.lua`, `auth.lua`.

Default Camera Settings

Defined in `driver.lua` via `CameraDefaultProps`:

- HTTP Port: `8080`
- RTSP Port: `8554`
- Main Stream path: `stream0`
- Sub Stream path: `stream1`
- Snapshot path: `tmp/snap.jpeg`

Initialization Flow

1. `OnDriverInit()` populates `_props` and initializes MQTT helper.
2. `InitializeCamera()` calls the remote init endpoint to fetch the public key.
3. `LoginOrRegister()` encrypts credentials (RSA-OAEP via external helper) and stores tokens.
4. `OnDriverLateInit()` pushes camera IP/port/auth to Camera Proxy and generates `Main Stream URL` / `Sub Stream URL` for the Control4 UI.

Notifications

- Events are processed by `HANDLE_JSON_EVENT()`; images are located via `GetImageForEvent()` and normalized with `normalize_http_url()`.
- Notification queue uses `NOTIFICATION_URLS` and `NOTIFICATION_QUEUE` to provide attachments to Control4.

Auth & MQTT

- `APPLY_MQTT_INFO()` requests broker details and credentials then calls `MQTT.connect()`.
- Tokens may be received via TCP; `UpdateAuthToken()` stores them and updates properties.

Maintainer Notes

- Remove camera-specific properties from `driver.xml` if you want the driver to rely solely on `CameraDefaultProps`.
- Keep `transport.execute(req, callback)` usage consistent when editing network calls.
- Use `normalize_http_url()` when decoding signed/escaped URLs from API responses.

Quick Dev Commands

```
git status
git add Smart-Camera-K26/driver.lua Smart-Camera-K26/README.md
git commit -m "Fix driver runtime errors; rely on CameraDefaultProps; update README"
```

Dynamic IP Auto-Update (Online Event Handling)

The driver includes intelligent handling of camera connectivity to ensure the correct IP address is always used.

Whenever the camera comes online, the driver automatically:

- Detects the **Camera Online** event
- Calls the **Get Devices API**
- Matches the device using **VID (Virtual ID)**
- Retrieves the latest **local IP address**
- Updates the **IP Address** property in the driver
- Pushes the updated address to the **Control4 Camera Proxy**

Events

The driver triggers Control4 events for automation:

Event ID	Event Name	Description
1	Motion Detected	Camera detected motion (alert, with snapshot)
2	Human Detected	Person identified in frame
3	Stranger Detected	Unknown face detected
4	Camera Online	Camera came online
5	Camera Offline	Camera went offline
6	Camera Restarted	Camera rebooted
7	Battery Low	Battery below 20%

Event Automation Examples

Use Control4 programming to:

- Turn on outdoor lights when motion detected
- Send push notification on human detection
- Display snapshot on TVs when stranger detected
- Alert when battery low
- Trigger security system on line crossing

Troubleshooting

Camera Not Waking Up

Check:

1. Camera has sufficient battery charge (check **Battery Level** property)
2. Solar panel receiving adequate sunlight
3. Camera is online and reachable on network
4. SDDP discovery working (test with **Discover Cameras (SDDP)**)

Action:

1. Run **Wake Camera (SDDP)** action
2. Wait full 7 seconds before accessing stream
3. Check battery level — charge if below 20%
4. Verify camera online status

No Video Stream

Check:

1. Camera is awake (7-second wake delay required)
2. RTSP port 8554 is open and accessible
3. Main/Sub Stream URL properties are populated
4. Battery level is sufficient (above 20%)
5. Network bandwidth is adequate

Action:

1. Run **Wake Camera (SDDP)** first
2. Wait 7 seconds
3. Run **Test Main Stream** or **Test Sub Stream**
4. Check generated RTSP URL in logs
5. Verify network connectivity to camera

Camera Not Initializing

Check:

1. IP address is correct and camera is reachable
2. Account credentials are valid
3. VID matches the actual device
4. Camera is powered on (check battery level)
5. Wi-Fi connection is stable

Action: Run **Initialize** action and check logs.

MQTT Events Not Working

Check:

1. **Enable MQTT** property is set to **True**
2. MQTT broker info is retrieved (Host, Port, Client ID, Secret)
3. Port 8884 is accessible
4. Network firewall allows MQTT over SSL

Action:

1. Run **Get MQTT Info** to populate MQTT settings
2. Run **Connect MQTT** to establish connection
3. Check **MQTT Host** and **MQTT Port** properties are filled
4. Verify in logs: `MQTT Connected`

Battery Draining Too Fast

Check:

1. Solar panel receiving direct sunlight (4+ hours/day minimum)
2. Too many streaming requests (each wake drains battery)
3. Event interval too short (increase from 5000ms)
4. MQTT generating too many events

Solution:

1. Reposition camera for better solar exposure
2. Reduce unnecessary streaming
3. Increase event interval to 10000ms or higher
4. Use HTTP polling instead of MQTT if too many events
5. Disable info notifications, keep only alerts

Snapshot Not Capturing

Check:

1. Camera is awake (run **Wake Camera (SDDP)** first)

2. HTTP Port 8080 is accessible
3. Authentication credentials are correct
4. Battery level sufficient

Action:

1. Wake camera and wait 7 seconds
2. Run **Test Snapshot** action
3. Check **Last Snapshot URL** property
4. Test URL in browser: `http://[YOUR_CAMERA_IP]:8080/tmp/snap.jpeg`

Discovery Not Finding Camera

HTTP Scan:

- Ensure camera IP is within scan range
- Adjust **IP Scan Range End** property
- Camera must respond to HTTP on port 8080
- Camera may be asleep (use SDDP instead)

SDDP:

- Camera must support SDDP protocol (K26-SL does)
- Multicast must be enabled on network
- Check firewall allows multicast traffic
- SDDP also wakes sleeping cameras

Authentication Failures

Check:

1. **Account** property has valid CldBus email
2. **Public Key** is retrieved from API
3. **Base API URL** is correct: `https://api.arpha-tech.com`
4. Network can reach API endpoint

Action:

1. Run **Login/Register** action
2. Check for `Authentication successful` in logs
3. Verify **Temp Token** property is populated

Camera Shows Offline

Check:

1. Battery is charged (check **Battery Level** property)
2. Solar panel working (check physical condition)
3. Wi-Fi signal strength is sufficient
4. Network has not changed (SSID/password)
5. Camera hasn't been reset

Solution:

1. Charge battery via USB if solar insufficient
2. Verify Wi-Fi connection on camera
3. Run **Get Camera Properties** action
4. Check **Online** property in driver

Solar Panel Not Charging

Check:

1. Panel clean and unobstructed
2. Direct sunlight exposure (not shaded)
3. Panel cable connected properly
4. Battery not damaged (check voltage)

Solution:

1. Clean solar panel surface
2. Reposition camera for better sun exposure
3. Verify physical connections
4. Replace battery if damaged

Technical Notes

RTSP URL Format

The driver generates RTSP URLs in the format:

```
rtsp://[IP]:[PORT]/streamtype=[TYPE]
```

- **IP:** Camera IP address
- **PORT:** RTSP port (default 8554)
- **TYPE:**
  - `0` = Sub stream (low quality, low bandwidth)
  - `1` = Main stream (high quality, higher bandwidth)

Example URLs:

```
Main: rtsp://[YOUR_CAMERA_IP]:8554/streamtype=1
Sub:  rtsp://[YOUR_CAMERA_IP]:8554/streamtype=0
```

CldBus API Authentication Flow

1. **Initialize Camera:** GET `/v1/init` → Retrieve RSA public key
2. **Login/Register:** Encrypt account with RSA-OAEP-SHA256 → POST `/v1/LoginOrRegisterUser` → Get temporary token
3. **Get Temp Token:** Generate HMAC-SHA256 signature → GET `/v1/TempTokenGet` → Verify temp token
4. **Exchange Token:** POST `/v1/TempTokenExchange` → Get exchange token
5. **Bind Device:** POST `/v1/BindDeviceToUser` → Associate VID with account
6. **Ongoing API Calls:** Use `Authorization: Bearer [auth_token]` header with HMAC signature on each request

MQTT Event Format

Events received on MQTT topic: `{VID}/events`

Motion Detection Example:

```
{
  "event": "motion",
  "timestamp": 1708300000,
  "snapshot_url": "http://[CAMERA_IP]:8080/snapshot/12345.jpg",
  "device_id": "[YOUR_DEVICE_VID]",
  "battery_level": 75
}
```


Human Detection Example:


```
{
  "event": "human",
  "timestamp": 1708300000,
  "snapshot_url": "http://[CAMERA_IP]:8080/snapshot/12346.jpg",
  "device_id": "[YOUR_DEVICE_VID]"
}
```


Battery Alert Example:


```
{
  "event": "battery_low",
  "timestamp": 1708300000,
  "battery_level": 15,
  "device_id": "[YOUR_DEVICE_VID]"
}
```


Connection Types

The driver maintains multiple connections:

1. **Camera Connection (5001):** Type: CAMERA, Facing: 6 (bidirectional), Purpose: Control4 camera proxy interface, Hidden: True
2. **Network/MQTT Connection (6001):** Type: TCP, Ports: 3333 (API), 8554 (RTSP), 8884 (MQTT SSL), Purpose: Network communication
3. **Keep-Alive Connection (7001):** Type: TCP, Port: 8081, Features: Auto-connect, keep-alive, connection monitoring, Purpose: Persistent connection to camera

Battery & Power Management

Solar Charging:

- Built-in solar panel continuously charges battery
- Requires 4+ hours direct sunlight per day
- Optimal: 6–8 hours full sun exposure
- Battery: Rechargeable lithium 5000mAh (typical)

Wake-on-Demand:

- Camera sleeps when inactive to save battery
- 7-second wake delay when accessing
- Auto-wake on streaming requests
- Auto-wake on initialization
- SDDP wake command supported

Battery Conservation Tips:

- Minimize streaming frequency
- Use sub stream instead of main stream
- Increase event interval (10000ms+)
- Disable info notifications (keep only alerts)
- Ensure good solar panel exposure

Encryption Level 2

Driver uses encryption level 2 for source code protection:

- `<devicedata encryption="2">` in driver.xml
- `<script file="driver.lua" encryption="2"/>` in driver.xml
- Protects driver.lua source code in .c4z package
- Prevents unauthorized modification

Video Codec Support

- **H.264 (Recommended):** Widely compatible, lower bandwidth
- **H.265 (HEVC):** Better compression, higher quality, requires more CPU

Control4 prefers H.264 for maximum compatibility.

Integration Examples

Example 1: Motion-Activated Lights

Scenario: Turn on outdoor lights when motion detected

```
When: Slomins K26-SL → Motion Detected
Then:
  - Outdoor Lights → Turn On
  - Wait 5 minutes
  - Outdoor Lights → Turn Off
```

Example 2: Security Alert on Human Detection

Scenario: Alert when person detected at night

```
When: Slomins K26-SL → Human Detected
If:   Time is between 10:00 PM and 6:00 AM
Then:
  - Send notification "Person detected outside"
  - Turn on: All Outdoor Lights
  - Start recording: NVR Camera Group
  - Send snapshot to mobile app
```

Example 3: Battery Low Alert

Scenario: Notify when battery needs attention

```
When: Slomins K26-SL → Battery Low
Then:
  - Send notification "K26 camera battery low - check solar panel"
  - Log: System event
```

Example 4: Perimeter Breach

```
When: Slomins K26-SL → Line Crossing
If:   Security System → State is "Armed"
Then:
  - Send notification "Perimeter breach detected"
  - Turn on: Perimeter Lights
  - Start recording for 5 minutes
  - Display snapshot on security panel
```

Support

Slomins Support:

- Website: www.slomins.com
- Email: support@slomins.com
- Phone: Customer service hotline

CldBus API Issues:

- API Endpoint: https://api.arpha-tech.com
- Contact: API support via Slomins

Control4 Integration:

- Requires Control4 Composer Pro for installation
- Compatible with Control4 OS 3.3.2 and newer
- Camera automation programming available

Building the Driver

Files Structure

```
Slomins-outdoor-K26/
├── driver.lua             # Main driver logic
```

```
├── driver.xml              # Configuration & metadata
├── mqtt_manager.lua        # MQTT event streaming & broker management
├── README.md               # This documentation
├── CldBusApi/              # API helper libraries
│   ├── auth.lua
│   ├── dkjson.lua
│   ├── http.lua
│   ├── sha256.lua
│   ├── transport_c4.lua
│   └── util.lua
└── build-c4z.ps1           # Build script
```

Build Package

Run in PowerShell:

```
.\build-c4z.ps1
```

This creates `Slomins-outdoor-K26-v1.1.0.c4z` ready for installation.

Manual Build

```
$files = @(
    "driver.lua",
    "driver.xml",
    "mqtt_manager.lua",
    "README.md",
    "CldBusApi\*.lua"
)

Compress-Archive -Path $files -DestinationPath "temp.zip"
Rename-Item "temp.zip" "Slomins-outdoor-K26.c4z"
```

Version History

v3.0 (Current)

- Added Face Detection: Distinguish between registered users and strangers
- Added "Registered User Detected" event (fires when a named face is detected)
- Fixed version display showing 0 in Control4 Manage Drivers interface

v1.0.7

- Implemented functionality to store all incoming events in the database.

v1.0.6

v1.0.6 (Current)

- Fixed device specific conditionals labels not showing in Control4 Composer

v1.0.5

- Removed hardcoded AppName

v1.0.4

- Updated Control4 Camera Programming Options
- **Device Events**: Added "Registered User Detected"
- **Commands Added**: Take Snapshot, Initialize Camera *, Unmute Mic, Mute Mic, Turn up speaker volume, Turn down speaker volume, Sensitivity =
- **Conditions Added**: If sensing motion, If not sensing motion, If Mic is muted, If Mic is unmuted, If Speaker volume is 1-10, If battery is <, >, or =, Sensitivity =

v1.0.3

- Send Control4 notification marked as critical if battery percentage is less than 15%
- Send notification every 4–6 hours (interval should be configurable in properties; default is 6 hours)
- Disable the timer if battery level is greater than 80%
- Enable the timer only when the first notification below 15% is received

v1.0.2

- convert the readme.md file to readme.html file

v1.0.1

-

v1.1.0

- CldBus API integration (api.arpha-tech.com)
- RTSP streaming support (streamtype=0/1 format)
- Battery management and monitoring
- Solar power support
- Wake-on-demand (7-second delay)
- MQTT event streaming over SSL (port 8884)
- Real-time detection (motion, human, stranger)
- SDDP discovery and wake command
- HTTP polling mode
- Snapshot capture with wake support
- PTZ controls with presets
- Encryption level 2 for driver protection
- Multi-device support via VID

License

Appendix: Quick Reference

Default Ports

- **HTTP:** 8080
- **RTSP:** 8554
- **TCP:** 3333, 8081
- **MQTT:** 8884 (SSL)

Example Device

- **Model:** K26-SL (solar_box_cam)
- **IP:** Assigned by your router (DHCP)
- **VID:** Unique per device
- **Wake Delay:** 7 seconds

RTSP URLs

```
Main: rtsp://[YOUR_CAMERA_IP]:8554/streamtype=1
Sub:  rtsp://[YOUR_CAMERA_IP]:8554/streamtype=0
```

Snapshot URL

```
http://[YOUR_CAMERA_IP]:8080/tmp/snap.jpeg
```

Primary Events

- **Motion Detected** — Movement in frame
- **Human Detected** — Person identified
- **Stranger Detected** — Unknown person
- **Battery Low** — Battery below 20%
- **Camera Offline** — Connectivity lost

Real-Time Event Detection

The driver supports **MQTT-based real-time event streaming over SSL (port 8884)** and provides notifications for:

- Doorbell ring notifications with snapshot
- Motion detection with snapshot attachment
- Human detection
- Face detection
- Stranger detection
- Camera online/offline status monitoring

MQTT Configuration

Default Behavior

MQTT is **enabled automatically by the driver after authentication**. The driver will automatically:

1. Enable MQTT
2. Fetch MQTT credentials
3. Connect to the MQTT broker
4. Subscribe to camera event topics

You normally **do not need to enable MQTT manually**.

CldBus App Motion Detection Settings

To receive **Motion Detection** and **Human Detection** events, detection must be enabled in the **CldBus mobile app**.

Steps

1. Open the **CldBus App**

2. Select your **camera device**
3. Tap **Settings**
4. Navigate to `Settings → Motion Detection`
5. Under **Detection Type**, select one of the following:

Detection Type	Description
All Detections	All movement events will trigger notifications. This includes general motion detection.
Human Detection	Only human movement will trigger events. Non-human motion will be ignored.

Push Notification Configuration

1. Create Push Notification

1. Open **Composer Pro**
2. Navigate to `Agents → Push Notification`
3. Click **Add Notification** then enter a name

Configure the following:

Setting	Value
Category	Cameras
Severity	Info / Critical
Subject	Camera Event

Click **Save**.

2. Enable Snapshot Attachment

Edit the created notification and set:

```
Attachment Type = Snapshot URL
```

This allows push notifications to include the **camera snapshot image**.

Mapping Push Notifications in Programming

1. In **Composer Pro**, open **Programming**.
2. Select the **Camera Driver** from the device list.
3. In the **Events** section, choose the event you want (e.g. **Motion Detected**).
4. On the right-side panel, click **Push Notifications**.
5. From the dropdown list, select the notification you created.
6. Drag and drop the notification into the programming area or double-click the green arrow.

Notification Flow

```
Camera Event (MQTT)
      ↓
Driver Receives Event
      ↓
Driver Processes Event
      ↓
Control4 Event Triggered
      ↓
Push Notification Agent
      ↓
Mobile Notification Sent
      ↓
Snapshot Image Attached
```

Battery Tips

- Ensure 4+ hours direct sunlight daily
- Keep solar panel clean
- Minimize streaming frequency

- Use sub stream to save battery
- Monitor battery level regularly

End of Documentation